THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE FULL SYLLABUS PAPER SET 04

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Dropping dough vertically or at an angle changes the: 1

(a) type of flour **(b)** colour of oil
(c) mass of the pan **(d)** way the dough enters the oil

2. In the puri-puffing study, the time taken to puff up is: 1

(a) the question **(b)** something we keep fixed
(c) an opinion **(d)** something we measure

3. The scientist who developed oil-eating bacteria is: 1

(a) Edward Jenner **(b)** Ananda Mohan Chakrabarty
(c) Louis Pasteur **(d)** Robert Hooke

4. Which is a non-communicable disease? 1

(a) Malaria **(b)** Measles **(c)** Chickenpox **(d)** Diabetes

5. Something we feel when unwell, like pain or tiredness, is a: 1

(a) symptom **(b)** sign **(c)** vector **(d)** pathogen

6. The incandescent lamp is an application of: 1

(a) magnetic effect **(b)** chemical effect only
(c) heating effect of current **(d)** static charge

7. In a dry cell, the carbon rod with the metal cap is the: 1

(a) electrolyte **(b)** negative terminal
(c) container **(d)** positive terminal

8. Friction arises because of: 1

(a) magnetism **(b)** irregularities of surfaces in contact
(c) gravity only **(d)** smoothness of air

9. During the day, air over land rises because land: 1

(a) cools faster than water **(b)** stays cold
(c) heats faster than water **(d)** is wet

10. Wind speed is greater when the pressure difference is: 1

(a) smaller **(b)** negative **(c)** greater **(d)** zero

11. Which of the following can flow? 1

(a) iron only **(b)** solid only **(c)** ice only **(d)** liquid and gas

12. The black mass iron sulfide is not attracted by a magnet because it is a: 1

(a) element **(b)** alloy
(c) new substance with different properties **(d)** mixture still

13. Alloys are mixtures of: 1

(a) minerals only **(b)** a metal with other metals or non-metals
(c) liquids only **(d)** gases only

14. The mass of 1 cm^3 of water is about: 1

(a) 0.1 g **(b)** 100 g **(c)** 1 g **(d)** 10 g

15. Rough surfaces: 1

(a) do not reflect at all **(b)** also reflect light (diffuse reflection)
(c) only absorb **(d)** only refract

16. The time between successive spring equinoxes is the: 1

(a) sidereal year **(b)** lunar year **(c)** leap year **(d)** tropical year

17. Which practice helps soil health? 1

(a) burning crop residue **(b)** organic manure
(c) excess pesticides **(d)** monoculture

18. In the chain millet → mouse → snake → eagle, the tertiary consumer is: 1

(a) mouse **(b)** millet **(c)** snake **(d)** eagle

19. The atmosphere is made of: 1

(a) rocks **(b)** air **(c)** ice only **(d)** water

20. The fusion of male and female gametes is: 1

(a) germination **(b)** regeneration **(c)** pollination **(d)** fertilisation

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. Mention two ways microorganisms are harmful. 2

22. Name two chronic diseases. 2

23. Why is an LED connected in a particular direction in a lemon cell? 2

24. Why do two balloons come close when we blow between them? 2

25. What are metalloids? Give two examples. 2

26. Why is floating ice important for aquatic life? 2

27. Why are mangroves in Sundarbans important? 2

28. Differentiate asexual and sexual reproduction in one point each. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Compare a fair test and an unfair test using an example. 3

30. Describe the activity that shows upthrust using an empty bottle. 3

31. Give three examples each of solids, liquids and gases. 3

32. Describe how you would identify a mirror by looking at your face in it from near. 3

33. Why were star positions at sunset, such as Makar near the winter solstice, recorded in ancient Indian texts like the Surya Siddhanta? 3

34. State why reuse, repair and recycle are important. 3

35. Write three safety rules while performing activities with electric circuits. 3

OR

Explain why even a ‘dead’ battery must not be thrown in garbage. 3

36. Name any three protected areas of India and their states. 3

OR

What would happen if all decomposers disappeared? 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. Explain with examples how scientific heritage of India (Ayurveda, variolation) contributed to health ideas and what modern science adds. 5

OR

A village has repeated cases of typhoid and dengue. Suggest preventive steps for each and say how the community can help. 5

38. Describe the formation of a cyclone and its structure. 5

OR

Compare a thunderstorm and a cyclone on formation, location, effects and duration. 5

39. A cold fizzy drink is opened in a hot place. Explain what happens with solubility of gases. 5

OR

Why do some objects float and others sink? Explain with density and give examples of wood, iron and ice. 5

40. Compare terrestrial and aquatic ecosystems with examples, abiotic factors and organisms. 5

OR

Describe how a change in one component, such as a drought, affects the entire ecosystem. 5

END OF QUESTION PAPER — Total: 80 Marks

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